

Insights from the Participatory Design of a Virtual Reality Spatial Navigation Application for Veterans with a History of TBI

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Figure 1: Snapshots from the VR application. Top: city, town, and airport environments. Bottom: the lobby area with progress displays and challenge coins; a collectible and proximity highlight; a destination helper arrow; and the wrist display showing user progress in real time.

ABSTRACT

We present a virtual reality (VR) application for potential application to cognitive rehabilitation of Veterans with a history of traumatic brain injury (TBI). Our application offers exploration of various virtual environments. We discuss the notable aspects of our design and implementation and insights from the participatory design process with lived-experience Veteran consultants.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Traumatic brain injury (TBI) can affect how the brain works, including cognition. In the U.S., an estimated 1.7 million people suffer from TBI, and 185,000 Veterans are diagnosed with TBI [1]. Environmental enrichment can improve symptoms in animal models [2]. We have designed and implemented a VR application that affords spatial navigation in novel environments as a potential application for cognitive rehabilitation in Veterans with a history of TBI. This work derives from our work on cognitive rehabilitation in brain-injured rodents exposed to novel environments [3]. We hypothesize that novel environments may activate neural populations important to spatial learning, which

could facilitate cognitive rehabilitation.

2 SYSTEM

Our VR application consists of five virtual environments built in Unity: mall, city, town, airport, and junior college (Figure 1). It runs on the Meta Quest 2 headset. We designed the environments to be inviting and non-threatening to avoid re-immersion or flashbacks for the users that could limit potential benefits in terms of cognitive rehabilitation. The user's goal is to explore the environment for familiarization, and then navigate to assigned destinations (e.g., Bookstore in the mall). Each environment consists of 16 levels designed with increasing difficulty in terms of area, number of spots, and other challenge-inducing aspects. In each level, destination and filler spots are generated randomly. After the user explores 80% of the area, a destination goal is presented. The user records arrival at the destination by pulling the trigger on the controller. Highlights were included on the ground to help with proximity estimation (Figure 1). If the user struggles to explore the environment or arrive at the destination within a pre-defined duration, helper arrows appear. The game can be played with a single controller in a standing or sitting position to accommodate mobility and limb-related disorders. A virtual hand is displayed throughout application use, and a wrist display can be activated by rotating the wrist inward towards the headset (Figure 1). Exploration percentage of the area and tasks are shown on the wrist display. Continuous steering locomotion is controlled with the thumbstick. Multimodal feedback is given (haptic, audio, and visual highlights) for important actions. Strategic landmarks aid users with their spatial navigation mapping (e.g., barn and windmill in the town environment in Figure 1). In the city and town environments, street names,

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consisting of words or numbers, were randomized between levels. No bird's-eye map was included to ensure that the user creates their own representation of the environment through exploration. In the higher levels, stationary (e.g., trash bin) and moving elements (e.g., non-player characters -NPCs, vehicles) increase the challenge and provide environment complexity (Figure 1). NPCs were designed with a full spectrum of skin color, facial features, height, and weight. Another challenge-inducing feature is degree of similarity, which includes altering features of filler spots, such as adding a name suffix (e.g., Cafe Tropical vs. Cafe Tropical Express), changing the sign color, adding wall stickers (e.g., 50% off), and changing the wall textures. Challenge is established by including similar filler spots in the environment. Each environment includes a background sound and various local 3D sounds (e.g., water near a fountain). These feature complexities accumulate as the levels progress.

The difficulty of the task is also adjusted based on the user's performance. Typically, in a level, the tasks include one, two, and three destinations (there are three tasks in each level). If the user fails at a task (by submitting an incorrect destination or not reaching the correct destination in a pre-defined time), the number of destinations is not increased for the upcoming task.

There is a lobby area that includes 3D environment icons, shelves, billboards, and information displays (Figure 2). Initially, the user completes a tutorial in the lobby and is informed about the anticipated benefits from the application (in alignment with [4]). Gamification elements include collectibles, challenge coins, and score. Collectibles appear near some destinations, are relevant to the destination, and are stored on the shelves in the user's lobby. Challenge coins are awarded for achievements, such as playing for fifteen minutes or completing a level with three stars. The score is calculated based on task success, path efficiency, and idle time above a threshold. Big numbers were included (out of 100,000), and the score is displayed on a large scoreboard in the lobby, following the best practices shared in [4]. We collect several metrics to evaluate user performance, including path taken, closeness to the optimal path, task correctness, U-turns, helper arrows, duration, walking distance, exploration percentage, stops, and level metrics (e.g., moving elements etc.).

3 INSIGHTS FROM PARTICIPATORY DESIGN

Four Veterans participated in the design of the VR application as lived-experience consultants (LECs). Each LEC playtested the application for approximately 10 hours of total VR exposure. Notable feedback was on motion sickness related to locomotion, adding familiar elements, gamification aspects for motivation, signifiers, and ease of mental association.

3.1 Motion Sickness Related to Locomotion

As this project focused on spatial exploration, a continuous steering locomotion technique was necessary. Several methods were tested by our LECs (steering only in the head's direction, steering with strafing, and steering in all directions), and steering in the head's direction was selected as the least motion sickness-inducing. A method to rotate other than physical body rotation was needed to cater to playing while sitting. We introduced instantaneous 30-degree snap rotation to reduce head displacement movement. Other motion sickness remedies were also tested (translational and rotational vignettes, and spatial frames of reference -a virtual display and a virtual nose). After testing, the LECs preferred the translational vignette and virtual nose, mentioning less subjective motion sickness with those elements incorporated. Although we utilize continuous steering locomotion, with these measures, the LECs can playtest for long hours without complaint.

3.2 Including Familiar Elements and Signifiers

The LECs mentioned that it would increase their engagement and motivation if the application had some military overtones and personal connections. Based on this, the user selects a military branch at the beginning and the lobby represents their branch. We included challenge coins and NPCs with military clothing. For a personal connection, we added the LECs' names as spot names (e.g., Sari's Place), which can be extended to incorporate any user's name – or name list – as a destination name. The LECs mentioned that similarity to real life would make the application more engaging. Modifications included spot and street names with numbers and NPC clothing appropriate for the different environments (e.g., city vs. town attire). We excluded a crying baby sound as the LECs stated it could be triggering for Veterans. The LECs mentioned having difficulty understanding how close they were to a destination when submitting a destination response. Hence, proximity highlights were added on the ground. When the helper arrows appeared before the LECs thought they needed help, they found it disheartening. Fine-tuning the timing of the helper arrows is expected to promote engagement.

3.3 Increasing Motivation

LECs mentioned liking the collectibles, especially their relevance to the visited spots, and seeing them in the lobby as a reminder of their progress. They mentioned being curious about each other's scores and how their scores compared. Although the potential benefit of the application was presented when setting up the application, they advocated for reminders of the anticipated long-term benefit as motivation, considering the long times spent in the application. They mentioned the variety in the environments helped keep their interest (e.g., distinct storefronts or buildings), as well as upbeat music with variety. A custom playlist may be considered.

3.4 Ease of Mental Association

The LECs identified difficulties in remembering layouts with similar types of buildings in a level (various healthcare spots). Distinguishing features on some of the buildings (e.g., the bread sign on the bakery spot in the town environment in Figure 1) helped LECs remember the layout. The specific spot names created confusion when they began with the same letter (e.g., Celestial Tower and Crystal Plaza). Hence, lists of names are continually refined to optimize novelty and map building.

ACKNOWLEDGMENT

The authors would like to thank the Department of Veterans Affairs for funding this project; the LECs Kevin Kupferer, Norman Spector, Sari Perlow, and Terry Stiff for taking part in the participatory design of this project; and Arya Mayfield, who worked as a student worker in the initial stages of this project.

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